

Latent Quality Risk in Pharmaceutical Manufacturing

A facility's quality risk is not directly observed. Regulators infer it from an intermittent record: periodic inspections graded on a three-point scale, and recall events. A raw inspection outcome bundles two things that inspection alone cannot separate — the underlying condition of the facility, and whether that particular inspection looked where the problems were.

Signal Structure of the Estimation Panel (12,888 plants, 2,693,592 plant-months)

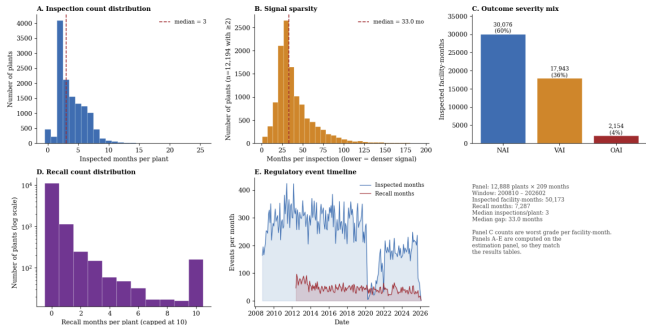


Figure 1. Estimation panel: 12,888 facilities, 209 months (Oct 2008–Feb 2026), 2,693,592 facility-months.

The record is sparse and coarse

- 50,173 inspected months; 7,287 recall months
- Under 2% of facility-months carry any signal
- Median facility: 3 inspections; median gap 33 months
- 60% NAI, 36% VAI, 4% OAI

What this requires

A monthly estimate from signals this thin requires an explicit assumption about how risk evolves between observations. A state-space formulation supplies one: risk drifts gradually, each inspection or recall is a noisy categorical draw from it, and months with no signal update nothing except the uncertainty around the estimate.

Method 1 — Probit with maximum likelihood (reference specification)

Let $q_{i,t}$ be the latent quality risk of facility i in month t . Observed outcomes enter through a latent index. For an inspection with cutoffs $T_1 < T_2$:

$$\Pr(\text{NAI}) = \Phi(T_1 - q_{i,t}), \quad \Pr(\text{VAI}) = \Phi(T_2 - q_{i,t}) - \Phi(T_1 - q_{i,t}), \quad \Pr(\text{OAI}) = 1 - \Phi(T_2 - q_{i,t}),$$

and for a recall, with cutoff T_3 : $\Pr(\text{recall}) = 1 - \Phi(T_3 - q_{i,t})$. Inspections are observed only in months when one occurs; the recall channel is observed every month. The latent path must be smoothed or it is unbounded. The period-to-period change $\theta_{i,t} = q_{i,t} - q_{i,t-1}$ is modeled as normal and converted to a probability using a tolerance parameter ξ :

$$\Pr(\theta_{i,t}) = \Phi\left(\frac{\theta_{i,t} + \xi}{\sigma}\right) - \Phi\left(\frac{\theta_{i,t} - \xi}{\sigma}\right).$$

Estimation maximizes the joint likelihood over all facility-months. Here μ , σ , ξ are hyperparameters, not estimates: σ scales the change probabilities, and is derived from the time scale rather than fitted.

Algorithm 1. Probit + MLE, coordinate ascent

- 1 **Input:** observations $y_{i,1:T}$, $i = 1, \dots, N$; fixed smoothing s.d. σ ; $T_1 = 0$.
- 2 Initialize latent states $q_{i,t}$ and cutoffs (T_2, T_3) .
- 3 **for** round $r = 1$ **to** R **do**
- 4 **for** each facility $i = 1, \dots, N$ **do**
- 5 maximize over q_i the outcome log-likelihood plus the change log-likelihood, by L-BFGS-B;
- 6 **end**
- 7 update cutoffs $(T_2, T_3) = \arg \max \sum_i L_i$ at the current states;
- 8 **end**
- 9 **Return:** point estimates $\hat{q}_{i,t}$ — full-sample by construction; no variance, no real-time series.

Method 2 — Laplace posterior-mode filter with RTS smoother

The same model as a state-space system. Transition: $q_{i,t} = q_{i,t-1} + \eta_{i,t}$, $\eta \sim N(0, \sigma_\eta^2)$; observation equations as above. Inference is sequential rather than joint. At each observed month the filtered estimate is the mode of the exact posterior,

$$\mu_{t|t} = \arg \max_q \left[\log N(q; \mu_{t|t-1}, P_{t|t-1}) + \log p(y_t | q) \right], \quad P_{t|t} = \left(-\frac{\partial^2}{\partial q^2} \right)^{-1} \Big|_{\mu_{t|t}}$$

a one-dimensional, log-concave problem with a unique solution. A Rauch–Tung–Striebel pass then runs backward: $\mu_{t|T} = \mu_{t|t} + G_t(\mu_{t+1|T} - \mu_{t+1|t})$, $G_t = P_{t|t}/P_{t+1|t}$.

Algorithm 2. Laplace posterior-mode filter + RTS smoother

- 1 **Input:** starting values and bounds for $\theta = \{\sigma_\eta, T_2, T_3\}$, $T_1 = 0$; observations $y_{i,1:T}$.
- 2 **while** the outer optimizer has not converged **do**
- 3 propose θ ; set $L = 0$;
- 4 **for** each facility $i = 1, \dots, N$ **do**
- 5 forward pass: predict $P_{t|t-1} = P_{t-1|t-1} + \sigma_\eta^2$; if an observation is present, set $\mu_{t|t}$ to the posterior mode by bounded search and $P_{t|t}$ to the inverse curvature there;
- 6 backward RTS pass for $(\mu_{t|T}, P_{t|T})$;
- 7 accumulate the marginal log-likelihood by Gauss–Hermite quadrature;
- 8 **end**; return $-L$ to the optimizer;
- 9 **end**; then a final forward and backward pass at θ^* .
- 10 **Return:** θ^* ; *filtered* $(\mu_{t|t}, P_{t|t})$ and *smoothed* $(\mu_{t|T}, P_{t|T})$ for every facility-month.

Method 1 has no filtered analogue. This distinction organizes the results that follow.

What the reformulation establishes

Numerical fidelity. The implementation reproduces the reference estimator exactly (maximum state difference 0.00e+00); the Gaussian approximation was verified against a bootstrap particle filter on 300 facilities (median correlation 0.996); and the two specifications agree closely on the estimated states ($r = 0.923$).

The smoothing parameter, estimated rather than assumed. Maximum likelihood returns $\sigma_\eta = 0.0429$, reproducing the value of approximately 0.048 derived independently from the time scale. The cutoffs likewise converge to similar values under both specifications despite different objectives.

Measurement and forecasting are different quantities. Scoring each held-out inspection by the state entering it:

Series	Information used	AUC	Concentration in top decile
Probit + MLE	full sample	0.953	8.6×
Filter, smoothed	full sample	0.928	6.7×
Filter, filtered (causal)	prior only	0.759	3.5×

Held-out later period, $n \approx 20,260$ inspections, base rate 5.1%.

The gap between the first two rows and the third is the value of hindsight: the full-sample series use information from after the month being scored. Only the filtered series is admissible as a forecast, and the filtering formulation is what makes it available at all.

What this does not establish. On next-inspection prediction specifically, simple summaries of a facility's inspection history perform comparably to the latent state. The implied probabilities are not calibrated and should be read as a ranking. And a record this sparse cannot tell us whether the score anticipates deterioration *within* a facility, as distinct from separating facilities that run persistently riskier from those that do not.

External validation: records the estimator never observed

Four FDA record types were withheld from estimation entirely. For each, we ask whether the real-time (filtered) state anticipates a *first* occurrence, restricting to facilities with no occurrence of that signal in the preceding twelve months, so that the test measures onset rather than the persistence of an existing problem. Evaluation is on a held-out later period.

Withheld record	Events	AUC (6 mo)	Top-decile concentration	Follows an OAI
Published 483s	1,091	0.778	4.3×	12.2%
Compliance actions	1,728	0.819	4.8×	32.1%
Import refusals	6,735	0.670	1.4×	1.3%
Inspection citations	11,660	0.628	1.6×	0.4%

Base rates 0.2%–3.1% across roughly 1.03–1.08 million facility-months. Baseline for the final column: 0.5%.

These four are not equally independent of the estimation inputs. Citations are recorded during inspections and 483s are issued when an inspection closes; compliance actions follow an adverse inspection finding roughly a third of the time. Import refusals are determined at ports of entry through a separate process, and are therefore the cleanest test of whether the score reflects facility quality rather than restating the inspection record it was built from. Refusals are anticipated at AUC 0.670 (95% CI [0.655, 0.683], resampled at the facility level), and the result is unchanged when restricted to months in which an inspection occurred (0.675) — the score is anticipating the outcome, not anticipating that an inspector arrives.

Interpretation. Given base rates well under one percent, concentration in the top decile is the meaningful quantity, not AUC. The evidence supports prioritization *across* facilities: identifying which facilities warrant attention among thousands. It does not support an early warning for any individual facility, and we do not claim one.